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Optimizing On-Site Search Engines in E-Commerce: A Data-Driven Approach to UX, AI, and User Behavior

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Abstract

On-site search engines constitute a critical component of e-commerce platforms, significantly influencing user experience and key performance indicators such as Click-Through Rate (CTR) and

Conversion Rate. This study examines the role and effectiveness of on-site search systems, aiming to evaluate their impact on product discovery and overall business performance.

The research combines an extensive literature review with the analysis of real-world usage data from a large e-commerce retailer in Greece. It investigates different search modalities (keyword-based search, filtering, and visual search), user behavior patterns (broad-to-narrow search strategies and filter usage), and underlying technologies, including Artificial Intelligence (AI), Natural Language Processing (NLP), and Learning to Rank methods. Additionally, key challenges such as zero-results queries, long-tail queries, and usability issues are explored.

The findings reveal the dominance of short queries, limited use of filtering mechanisms, frequent refinement of general searches, and the significant influence of the search bar on overall website traffic. These results both confirm and extend existing literature, highlighting behavioral patterns that remain underutilized in current system design.

The study concludes with practical recommendations for improving on-site search performance, including enhanced filtering functionality, improved usability, and the integration of advanced AI techniques. The contribution of this research lies in bridging theoretical approaches with real-world data, offering actionable insights into the optimization of e-commerce search systems.

1. Introduction

E-commerce has emerged as one of the most dynamic and rapidly evolving sectors of the global economy. The proliferation of digital technologies, the widespread adoption of the internet, and the transformation of consumer behavior have created a highly competitive environment in which user experience (UX) plays a critical role in determining business success (Gupta & Bansal, 2024). Within this context, on-site search engines—tools that enable users to locate products within an

e-commerce platform—have become a fundamental component of digital navigation, directly influencing product discovery, conversion rates, and overall customer experience (Makrydakís & Skopeteas, 2022).

Beyond their functional role, on-site search systems hold strategic importance. Prior research highlights that search is often the primary mode of interaction between users and large product catalogs, particularly in environments with extensive product assortments (Wen, 2011; Kent, 2014). Users frequently prefer search over menu-based navigation, as it provides a faster and more direct path to desired results (Zeebaree, 2025). Consequently, on-site search acts as a key driver within the customer journey, reducing information overload and facilitating decision-making.

Despite its importance, the effectiveness of on-site search remains constrained by several challenges. User behavior studies identify a common “broad-to-narrow” search pattern, where users begin with general queries (e.g., “laptop”) and progressively refine them through query reformulation or filtering (Yamin & Ramayah, 2011). However, the adoption of filtering mechanisms remains relatively low, often due to suboptimal interface design or limited perceived usefulness (Zeebaree, 2023; Saura et al., 2017). Furthermore, search systems frequently struggle with complex or long-tail queries, leading to irrelevant or zero-results pages that negatively impact user satisfaction and increase abandonment rates (Nguyen, 2020).

Recent technological advancements have significantly enhanced the capabilities of on-site search engines. Techniques such as Natural Language Processing (NLP), synonym expansion, spell correction, and Learning to Rank (LTR) models have improved the relevance and accuracy of search results (Usta et al., 2021; Zhou & Agichtein, 2020). Additionally, the integration of Artificial Intelligence (AI) technologies, including personalization and recommendation systems, has enabled more adaptive and user-centric search experiences (Lalitha & Sreeja, 2021; Ziakis & Vlachopoulou, 2023). Despite these advancements, a gap remains between theoretical developments and their practical implementation in real-world e-commerce environments.

This study aims to address this gap by combining a comprehensive literature review with an empirical analysis of real-world user behavior data. Specifically, it investigates how users interact with on-site search systems and how these interactions influence key performance indicators such as Click-Through Rate (CTR), Conversion Rate, and Average Order Value (AOV). The analysis is based on anonymized search data collected over a nine-month period from a large e-commerce retailer in Greece.

The research focuses on several critical aspects, including query characteristics, filter usage, refinement behavior, and the occurrence of no-results pages. It also examines the impact of design and technological interventions, such as visual elements, long-tail query support, and adaptive search mechanisms, on system performance and user behavior.

The contribution of this study is twofold. First, it provides empirical evidence on user search behavior within a real-world e-commerce context, highlighting patterns that are often underrepresented in the literature. Second, it bridges the gap between theoretical models and practical applications by offering actionable insights and recommendations for improving on-site search performance.

Ultimately, this research positions on-site search as a strategic tool for enhancing both user experience and business outcomes in e-commerce platforms.

2. Literature Review

2.1 Evolution of On-Site Search Engines

The evolution of on-site search engines has closely followed the broader development of information technologies and e-commerce systems, transitioning from simple keyword-based retrieval models to complex, AI-driven architectures. In their early stages, search systems primarily relied on keyword matching and Boolean operators, which, although functional, were limited to exact term matching and often resulted in irrelevant or incomplete outputs (Kent, 2014). This limitation negatively affected both user experience and system effectiveness.

Over time, the growing complexity of e-commerce platforms and user expectations necessitated the development of more advanced and context-aware search systems. The integration of Natural Language Processing (NLP) techniques enabled improved handling of synonyms, spelling variations, and diverse query formulations (Shaban & Zeebaree, 2023; Usta et al., 2021). A major milestone in this evolution was the introduction of Learning to Rank (LTR) algorithms, which leverage supervised learning techniques to optimize result ranking based on user interaction signals, such as clicks and engagement (Zhou & Agichtein, 2020). Furthermore, reinforcement learning approaches (e.g., RLIRank) have enabled dynamic adaptation of ranking strategies based on real-time user behavior (Usta et al., 2021).

The shift from static keyword-based systems to adaptive and personalized search solutions has been further supported by advancements in Big Data technologies and scalable infrastructures, allowing real-time processing of large-scale datasets (Lalitha & Sreeja, 2021). These developments have been essential for managing the increasing complexity of product catalogs and meeting rising user expectations in modern e-commerce environments (Makrydakis & Skopeteas, 2022). As a result, on-site search engines have evolved into flexible, data-driven systems capable of continuously adapting to user needs.

From a system design perspective, the development of effective e-commerce search engines requires careful consideration of architectural components such as indexing, data management, and scalability, in order to ensure reliable and high-performance retrieval (Li, 2010).

More recently, the rapid advancement of transformer-based models and Large Language Models (LLMs), such as BERT and GPT, has enabled deeper semantic understanding of user queries. Contemporary approaches leverage context-aware representations to bridge the vocabulary gap between user queries and product descriptions. For instance, Lu et al. (2021) propose an end-to-end product retrieval system based on Graph Convolutional Networks and multilingual BERT, which

incorporates contextual information from related queries to enhance product representation. Their approach demonstrates significant improvements in retrieval performance, including up to a 25% increase in recall and notable gains in business-related metrics such as click-through rate and conversion.

Overall, the literature highlights a clear transition toward intelligent, context-aware, and user-centric search systems. These advancements have redefined the role of on-site search, enabling systems not only to retrieve relevant results but also to interpret user intent, handle ambiguity, and support personalized experiences, thereby overcoming the limitations of traditional search models.

2.2 Search Types and Optimization Techniques

Search queries in e-commerce environments can be categorized based on both their structure and user intent. The literature commonly distinguishes between short-tail queries (e.g., “laptop”) and long-tail queries (e.g., “15-inch gaming laptop”), highlighting that while short queries are more frequent, they are less specific, whereas long-tail queries occur less often but are associated with higher conversion potential (Nguyen, 2020). Additionally, user behavior studies indicate that search interactions often follow a “broad-to-narrow” pattern, where users initially submit general queries and progressively refine them to achieve more precise results (Leung & Urmitsky, 2025; Penha et al., 2023).

The effective handling of such query types has driven the development of various optimization techniques aimed at improving retrieval performance. Common approaches include spell correction mechanisms and synonym handling, which enhance query understanding and reduce mismatches between user input and product data (Shaban & Zeebaree, 2023; Bastos, 2023). Furthermore, the application of Natural Language Processing (NLP) enables systems to capture user intent and contextual meaning, thereby reducing the occurrence of zero-results pages and improving result relevance (Usta et al., 2021).

Another key component of search optimization is the implementation of faceted search through filtering systems, allowing users to refine results based on attributes such as price, category, or brand (Makrydakis & Skopeteas, 2022; Kent, 2014). When effectively designed, these mechanisms enhance navigation efficiency and reduce cognitive load by structuring the search space.

Advancements in optimization techniques are also closely linked to the need for fast response times and accurate result ranking. Modern e-commerce search systems integrate multiple approaches, including semantic search, contextual understanding, and AI-driven re-ranking models, in order to meet the expectations of increasingly demanding users (Saura et al., 2017; Ziakis & Vlachopoulou, 2023; Lalitha & Sreeja, 2021). The adoption of such techniques is considered critical for maintaining competitiveness in digital marketplaces.

Emerging trends further extend the capabilities of search systems through the integration of visual content search. According to Dagan et al. (2023), visual search enhances user engagement by reducing cognitive effort and enabling a more exploratory, visually guided interaction with products. Visual elements, such as product images and banners, act as attention amplifiers, particularly for

users who may struggle to articulate precise queries. This approach appears to be especially prevalent among younger, digitally native users, such as Generation Z, highlighting a shift toward more intuitive and multimodal search experiences.

2.3 UX Design and User Behavior in On-Site Search

User experience (UX) in on-site search engines represents a critical determinant of success for e-commerce platforms. Search is not merely a technical functionality, but a process of interpreting user intent and delivering relevant results in an efficient and intuitive manner (Makrydakīs & Skopeteas, 2022; Bastos, 2023). The design of search interfaces encompasses multiple factors, including the visibility and placement of the search bar, autocomplete functionality, ease of filter usage, response speed, and the overall usability of the search environment (Kent, 2014; Saura et al., 2017).

Prior research indicates that users have high expectations from search systems, requiring fast, accurate, and contextually relevant results, along with support for spelling errors and natural language queries (Bastos, 2023; Zeebaree, 2023). Failure to meet these expectations can lead to user frustration and eventual abandonment of the search process (Nguyen, 2020). In particular, zero-results pages represent a critical usability issue, as they are strongly associated with negative user sentiment, reduced click-through rates, and increased bounce rates (Bastos, 2023).

User behavior analysis reveals consistent interaction patterns within search environments. One of the most prominent is the “broad-to-narrow” search strategy, where users initiate their search with general queries and subsequently refine them through query reformulation or filtering (Leung & Urminsky, 2025; Penha et al., 2023). Despite the importance of filtering mechanisms, their usage remains relatively low, often due to poor interface design or unclear labeling (Zeebaree, 2023). This finding contrasts with earlier studies suggesting that well-designed filters can significantly enhance navigation efficiency and reduce cognitive load (Saura et al., 2017).

Another important aspect of user behavior is query length. Most search queries in e-commerce contexts are short, typically consisting of one or two words, reflecting users’ expectation that systems will infer context and intent automatically (Zuo et al., 2022). This behavior underscores the importance of incorporating techniques such as synonym matching, semantic search, and spell correction to improve retrieval accuracy (Shaban & Zeebaree, 2023). At the same time, the limited use of filters often results in overly broad result sets, which can hinder decision-making and reduce search effectiveness (Kent, 2014).

Optimizing UX in on-site search extends beyond core retrieval performance and includes the integration of complementary navigation tools, such as category structures, visual elements, and recommendation systems. The inclusion of recommender systems within search interfaces has been shown to improve user engagement and guide users toward relevant products (Lalitha & Sreeja, 2021). Additionally, the use of visual elements, such as product images and inline filters, supports better comprehension of available options and facilitates decision-making.

Understanding user behavior is therefore essential for the design of effective on-site search systems. The ability to monitor user interactions—including clicks, query refinements, filter usage, and no-result events—provides valuable insights into user needs and system limitations (Bastos, 2023). Building on these insights, this study connects theoretical findings with empirical data, aiming to identify opportunities for improving search experience through targeted UX and system-level interventions.

2.4 Filtering Systems and Personalization

Filtering systems constitute a fundamental component of search architecture in e-commerce platforms, enabling users to refine search results based on specific criteria such as price, brand, technical attributes, or color (Makrydakis & Skopeteas, 2022; Kent, 2014). Despite their recognized importance, the use of filters remains relatively limited, with studies indicating that users often ignore or underutilize available filtering options due to poor interface design or low perceived usefulness (Zeebaree, 2023; Saura et al., 2017). Factors such as visual prominence, ease of multi-parameter selection, and clarity of labeling play a critical role in determining filter adoption (Bastos, 2023). The literature emphasizes the importance of faceted search, where users iteratively refine results based on predefined attributes, highlighting the need for flexible and context-aware filtering mechanisms (Makrydakis & Skopeteas, 2022).

Advanced approaches to filtering include dynamic facet generation, as proposed by Duan et al. (2013), where probabilistic models are used to identify and rank the most informative attributes for filtering. This approach enables adaptive filtering structures that respond to user queries in real time, improving both relevance and usability.

In parallel, the integration of personalization techniques has significantly transformed on-site search systems. Personalization allows search results and filtering options to be tailored to individual user preferences and browsing history (Lalitha & Sreeja, 2021; Tokuç & Dag, 2025). These techniques rely on machine learning models that analyze user behavior, including search queries, clicks, and purchase history, in order to generate dynamic product recommendations, personalized filters, and adaptive ranking strategies (Ziakos & Vlachopoulou, 2023). This approach is closely aligned with recommender systems, which are often integrated within search interfaces to enhance relevance and user engagement. Empirical evidence suggests that even relatively simple personalization strategies, such as reminding users of previously viewed products, can significantly improve conversion rates (Jannach & Ludewig, 2017).

However, the application of personalization introduces several challenges. Issues related to privacy and transparency are particularly critical, as users may be unaware of or uncomfortable with the extent to which their data are being analyzed (Saura et al., 2017). Additionally, excessive personalization may lead to the formation of “filter bubbles,” limiting exposure to diverse product options and potentially constraining user choice (Bastos, 2023).

The integration of filtering systems with personalization presents significant opportunities for enhancing user experience. By leveraging search data and AI-driven techniques, e-commerce platforms can deliver dynamic filters, personalized suggestions, and context-aware search experiences that better align with user intent. However, the effective implementation of such systems requires robust data infrastructure, advanced analytical capabilities, and continuous model

optimization based on user feedback and behavioral patterns (Usta et al., 2021; Zhou & Agichtein, 2020).

Finally, emerging approaches such as exclusion filters—implemented through SQL- or Python-based systems (Priya & Vijayalakshmi, 2025)—offer additional flexibility by allowing users to explicitly exclude unwanted products, brands, or attributes from search results. Such mechanisms enhance user control and may be particularly valuable for specific user segments with well-defined preferences, contributing to a more personalized and efficient search experience.

2.5 AI Technologies and Machine Learning Models in On-Site Search Engines

Early research on the design of production-level search systems, such as the work of Li (2010), focused on backend architecture and indexing strategies, establishing the foundations for the development of modern AI-powered search systems. The integration of Artificial Intelligence (AI), particularly Machine Learning (ML) and Deep Learning techniques, has since transformed the functionality and effectiveness of on-site search engines, enabling more advanced prediction, pattern recognition, and decision-making capabilities (Wang, 2024).

A key advancement in this domain is the adoption of Learning to Rank (LTR) algorithms, which move beyond simple keyword matching by ranking search results based on the estimated relevance or likelihood of user engagement (Usta et al., 2021; Zhou & Agichtein, 2020). These models leverage user interaction data—such as clicks, conversions, and dwell time—to continuously improve ranking performance and enhance result relevance.

LTR models are often combined with reinforcement learning approaches (e.g., RLIRank), allowing search systems to dynamically adapt to changes in user behavior and optimize result presentation in real time (Usta et al., 2021). This dynamic adaptation is closely related to personalization and contextual search, as systems learn from user interactions—including queries, clicks, and purchase behavior—to deliver more tailored search experiences (Santu et al., 2017). As a result, such approaches contribute to reducing no-results rates, increasing click-through rates, and improving overall user satisfaction.

In parallel, the application of Natural Language Processing (NLP) techniques has significantly enhanced the ability of search systems to interpret user intent. NLP enables functionalities such as synonym handling, spelling correction, query understanding, and natural language interaction (Shaban & Zeebaree, 2023). The use of semantic search and contextual embeddings further improves the relevance of search results, even in complex or ambiguous queries (Lalitha & Sreeja, 2021; Bastos, 2023). Additionally, the emergence of multimodal search approaches—integrating text, images, and video—has expanded the capabilities of modern search systems (Ziakos & Vlachopoulou, 2023).

Despite these advancements, the effective deployment of AI-driven search systems requires high-quality training data, continuous model updates, and rigorous performance monitoring through techniques such as A/B testing and real-world usage analysis (Bastos, 2023). At the same time, several challenges remain, including computational complexity, scalability requirements, and the trade-off between accuracy and response time (Kent, 2014; Gupta & Bansal, 2024).

Finally, the integration of AI-driven search with other platform components—such as recommendation systems and personalization layers—enables a more holistic user experience. This integration enhances user engagement, supports product discovery, and contributes to increased

customer retention and sales performance (Lalitha & Sreeja, 2021; Ziakis & Vlachopoulou, 2023). Recent approaches further explore the combination of large language models (LLMs) with cloud-based infrastructures to optimize recommendation performance, as demonstrated by Xu et al. (2024) through the UniLLMRec framework. Such developments highlight the ongoing convergence of AI, data infrastructure, and search technologies in modern e-commerce ecosystems.

2.6 Challenges and Limitations of On-Site Search Engines

Despite significant technological advancements and increasing system sophistication, on-site search engines continue to face critical challenges and limitations that directly impact user experience and the overall performance of e-commerce platforms. One of the most widely documented issues is the occurrence of zero-results pages, which negatively affect user satisfaction and reduce the likelihood of purchase completion (Bastos, 2023). Users often experience frustration when queries fail to return relevant results, particularly in cases involving synonyms, product combinations, or less common search terms, frequently leading to search abandonment and website exit (Nguyen, 2020).

Another major challenge concerns the handling of long-tail queries. Although less frequent, these queries are highly valuable as they reflect more specific purchase intent and are often associated with higher conversion rates (Makrydakis & Skopeteas, 2022). However, many search systems struggle to interpret and respond effectively to such queries due to limited training data or insufficient model generalization capabilities (Kent, 2014; Gupta & Bansal, 2024). This limitation can result in irrelevant outputs or zero-results pages, ultimately undermining user trust in the system.

A critical issue in AI-driven search systems, particularly those incorporating personalization and recommendation models, is the cold start problem (Wang et al., 2024). This phenomenon manifests in two forms: user cold start, where insufficient data exist for new users, and item cold start, where newly introduced products lack sufficient interaction history. Both cases hinder accurate ranking and personalization, reducing the effectiveness of the search experience during initial interactions.

Limitations also arise in the handling of multilingual queries and cultural variability. Many on-site search systems support a limited number of languages or fail to account for linguistic and cultural differences, thereby restricting global usability and accessibility (Ziakis & Vlachopoulou, 2023). Additionally, the increasing complexity of AI-driven models introduces scalability challenges, as processing large volumes of queries in real time requires optimized architectures and substantial computational resources (Usta et al., 2021; Zhou & Agichtein, 2020).

Response time represents another critical factor affecting search performance. Users expect near-instantaneous results, and even minor delays can negatively impact their experience (Bastos, 2023). Ensuring low latency requires not only efficient algorithms but also robust infrastructure, including optimized indexing, caching strategies, and load balancing mechanisms.

Furthermore, the integration of personalization and recommendation systems introduces concerns related to privacy and transparency. Users are often unaware of how their data are collected and utilized, which may lead to distrust and resistance toward personalized experiences (Saura et al., 2017). Balancing personalization with data protection and ethical considerations remains an ongoing challenge in modern e-commerce systems.

The literature also emphasizes the importance of continuous monitoring and evaluation of search system performance. Metrics such as click-through rate, no-results rate, and user interaction patterns are essential for identifying system weaknesses and guiding iterative improvements (Makrydakis &

Skopeteas, 2022; Usta et al., 2021). The absence of such practices can result in system stagnation and reduced adaptability to evolving user behavior.

Overall, the challenges and limitations of on-site search engines highlight the need for continuous technological investment, user-centered design, and responsible data management. Addressing these issues is essential for developing robust, scalable, and effective search systems that enhance both user experience and business performance in e-commerce environments.

2.7 Synthesis and Research Positioning

The literature review highlights the central role of on-site search engines as a key component of user experience in e-commerce environments. The evolution from basic keyword-based retrieval systems to advanced, AI-driven and personalized search architectures reflects the increasing complexity of digital commerce and the growing expectations of users (Wen, 2011; Kent, 2014; Makrydakis & Skopeteas, 2022). The adoption of techniques such as Learning to Rank (LTR), reinforcement learning approaches, Natural Language Processing (NLP), semantic embeddings, and multimodal models has significantly enhanced the ability of search systems to deliver relevant, accurate, and context-aware results (Usta et al., 2021; Zhou & Agichtein, 2020).

Despite these advancements, the literature consistently identifies several persistent challenges that limit system effectiveness. Issues such as zero-results pages, inadequate handling of long-tail queries, the cold start problem, multilingual limitations, and high computational requirements continue to affect both user experience and system performance (Bastos, 2023; Nguyen, 2020; Kent, 2014; Ziakis & Vlachopoulou, 2023). Additionally, low filter usage and UX design limitations further emphasize the need for continuous system optimization and user-centered design approaches (Zeebaree, 2023; Saura et al., 2017).

The literature also underscores the importance of integrating search functionality with other components of the e-commerce ecosystem, such as recommendation systems, filtering mechanisms, and visual elements, in order to create a cohesive and efficient user experience (Lalitha & Sreeja, 2021). At the same time, the increasing reliance on personalization introduces challenges related to privacy, transparency, and the potential formation of filter bubbles (Saura et al., 2017; Bastos, 2023).

Building on these findings, this study aims to bridge the gap between theoretical models and real-world implementation by examining the performance of an on-site search engine in a large Greek e-commerce retailer. Using anonymized search data, the analysis focuses on identifying key user behavior patterns, including query length, click-through rates (CTR), filter usage, refinement behavior, and abandonment associated with zero-results pages.

The objective of this research is to evaluate the extent to which theoretical insights are reflected in real-world search behavior, while also identifying deviations and previously underexplored patterns.

Based on these findings, the study seeks to provide practical recommendations for improving search performance, enhancing user experience, and supporting business outcomes in e-commerce environments. Ultimately, this research contributes to the literature by offering empirical validation of existing models and generating new insights that can inform future research on on-site search systems.

3. Methodology

3.1 Research Objectives

This study aims to investigate the role and impact of on-site search engines in e-commerce environments, combining both theoretical and empirical perspectives. The primary objective is to evaluate on-site search as a key driver of user experience and conversion performance.

Specifically, the study focuses on:

- analyzing user behavior during search interactions,
- examining the relationship between search features and key performance indicators (KPIs),
- and evaluating the impact of design and technological interventions on search effectiveness.

By addressing these objectives, the study seeks to provide a comprehensive understanding of how on-site search contributes to both user engagement and business outcomes.

3.2 Data and Study Period

The empirical analysis is based on real-world data collected from a large-scale Greek e-commerce retailer. The dataset covers the period from September 2, 2024, to May 31, 2025, providing a sufficiently broad temporal scope that captures seasonal variations while avoiding bias from isolated promotional periods.

All data were anonymized to ensure full compliance with the General Data Protection Regulation (GDPR). The dataset includes both quantitative and behavioral metrics, such as:

- search queries,
- user sessions,
- clicks,
- and conversion-related metrics.

Additionally, key performance indicators (KPIs) such as conversion rate (CR), refinement rate, and revenue attributed to search interactions were analyzed. This combination of behavioral and performance data enables a comprehensive evaluation of search system effectiveness.

3.3 Methodological Approach

The study adopts a descriptive and exploratory data analysis approach. The methodology is structured into two main stages:

1. **Descriptive Analysis**

Initial analysis focuses on identifying general trends and patterns in user behavior, including query frequency, session-level search activity, and overall search performance.

2. **Experimental Analysis**

A series of targeted experiments is conducted to evaluate the impact of specific variables on user behavior and system performance.

The analysis incorporates statistical methods, data visualization techniques, and computational tools to examine both quantitative metrics and behavioral patterns.

Key variables analyzed include:

- number of searches per session,
- revenue generated through search,
- proportion of queries leading to conversions,
- and contribution of search to total revenue.

In addition, qualitative behavioral indicators are examined, such as query specificity, use of natural language, interaction with filters, and behavior on search result pages.

3.4 Study Limitations

This study presents certain limitations that should be acknowledged. First, the analysis is based on data from a single e-commerce platform, which may limit the generalizability of the findings across different industries or markets. Second, the study does not incorporate qualitative research methods, such as interviews or focus groups, which could provide deeper insights into user motivations and decision-making processes.

Nevertheless, the scale and granularity of the dataset allow for robust conclusions regarding user behavior and search performance within a real-world e-commerce context.

3.5 Experimental Design

To further investigate the impact of search-related factors, four targeted experiments were designed and conducted:

- **Experiment 1:**
Examines the effect of enhancing the search results page (e.g., improved filters, introduction of tabs, and visual banners) on refinement behavior, conversion rate, and search-generated revenue.
- **Experiment 2:**
Investigates the role of visual content and promotional banners in influencing user

engagement and session duration, assessing the relationship between visual elements and performance metrics.

- **Experiment 3:**

Analyzes user preference for general queries and the adoption of broad-to-narrow navigation strategies. The experiment evaluates whether structured search result presentation supports incremental query refinement and improves search efficiency.

- **Experiment 4:**

Explores the increasing use of long-tail, natural language queries, potentially influenced by the widespread adoption of AI-driven interfaces. The analysis examines differences in query formulation, performance, and implications for NLP-enabled search systems.

Finally, the study includes structured datasets and aggregated tables (e.g., weekly and monthly summaries), enabling comparative analysis and supporting the interpretation of experimental results. These datasets form the foundation for the analysis presented in the subsequent section.

3.5.1 Experiment 1: Impact of Search Results Page Redesign on User Behavior and Performance

This experiment examines the relationship between the enrichment of the search results page (Search Display Page) and both user behavior and system performance.

The central hypothesis is that enhancing the search results page with functional and visual elements—such as filters, category tabs, and promotional banners—improves search effectiveness by reducing refinement behavior, increasing conversion rates, and generating higher revenue from search interactions.

The experimental design consists of two phases:

- Phase 1: Baseline measurement of search performance for a high-frequency query (“Samsung”) under a minimal interface configuration.
- Phase 2: Evaluation after redesign, including the introduction of visual elements, structured navigation (tabs), and enhanced filtering options.

Performance is assessed by comparing key metrics across the two phases, including refinement rate, conversion rate, and revenue contribution.

H₁: Enhancing search results pages with functional and visual elements improves the overall effectiveness of the search system.

3.5.2 Experiment 2: The Role of Visual Content and Promoted Banners in User Engagement and Performance

This experiment examines the relationship between the enrichment of the search results page (Search Display Page) and both user behavior and system performance.

The central hypothesis is that enhancing the search results page with functional and visual elements—such as filters, category tabs, and promotional banners—improves search effectiveness

by reducing refinement behavior, increasing conversion rates, and generating higher revenue from search interactions.

The experimental design consists of two phases:

- **Phase 1:** Baseline measurement of search performance for a high-frequency query (“Samsung”) under a minimal interface configuration.
- **Phase 2:** Evaluation after redesign, including the introduction of visual elements, structured navigation (tabs), and enhanced filtering options.

Performance is assessed by comparing key metrics across the two phases, including refinement rate, conversion rate, and revenue contribution.

H₁: Enhancing search results pages with functional and visual elements improves the overall effectiveness of the search system.

3.5.3 Experiment 3: Long-Tail Natural Language Queries and the Influence of AI on Search Behavior

This experiment explores the growing use of long-tail, natural language queries and their relationship with the increasing adoption of AI-driven interfaces.

The analysis is motivated by the observation that users increasingly formulate queries in a more descriptive and conversational manner (e.g., “laptop for remote work with long battery life”), reflecting interaction patterns influenced by AI tools.

The experimental approach involves identifying and analyzing long-tail queries within a defined time window and comparing their performance against traditional keyword-based queries across key metrics such as CTR, no-results rate, and conversion rate.

H₃: Increased user familiarity with AI technologies leads to greater adoption of long-tail, natural language queries, with measurable effects on search performance.

3.5.4 Experiment 4: Preference for Generic Queries and the “Broad-to-Narrow” Search Behavior

This experiment focuses on users’ preference for short, generic queries and their tendency to prioritize speed and immediacy during search interactions.

Users frequently initiate their search process using broad queries (e.g., “laptop”, “refrigerator”, “Samsung”) and subsequently refine results through filters or query reformulation. This behavior reflects a “broad-to-narrow” navigation strategy.

The analysis examines:

- the proportion of generic queries,
- subsequent user actions (filter usage, refinements),
- and conversion outcomes.

Additionally, differences across device types and product categories are explored.

H4: Users exhibit a strong preference for generic queries, prioritizing speed and flexibility in search interactions.

3.6 Data Structure and Analytical Tables

The datasets used in this study are presented through structured tables aggregated on a weekly and monthly basis. These tables include key performance indicators such as sessions, queries, click-through rate (CTR), conversion rate, and revenue metrics.

The tabular data are complemented by visualizations that facilitate the interpretation of trends and support the comparative analysis across experiments. These datasets form the empirical foundation for the results presented in the following section.

4. Results

4.1 Experiment 1: Impact of Search Results Page Enrichment on Search Performance

This experiment examines the effect of enhancing the search results page (Search Display Page) on user behavior and key performance indicators. The analysis focuses on the high-frequency query “Samsung”, which was recorded 6,435 times over two consecutive months.

During the first period (September 2024), the search results page included only basic filtering options and lacked structured visual guidance. In the second period (October 2024), the page was redesigned to include category tabs (e.g., Smartphones, Smartwatches, Laptops), promotional banners, and enhanced filtering functionality.

The comparative analysis reveals significant improvements across multiple performance metrics.

Μήνας	Queries (Samsung)	Refinement Rate (%)	Queries Led to Purchase	CR (%)	AOV (€)	Income from Samsung Search (€)
Σεπτέμβριος 2024	3.146	50	28	0.89	310	8.680
Οκτώβριος 2024	3.289	25	35	1.19	330	11.550

First, the **refinement rate decreased substantially**, from 50% to 25%. This reduction suggests that users were able to identify relevant products more efficiently, reducing the need for query reformulation or additional filtering. This finding aligns with prior research indicating that high refinement rates are often associated with suboptimal result presentation and increased cognitive

effort. The improved structure and visual organization of the results page appear to reduce search friction and support more direct decision-making.

Second, the **conversion rate increased** from 0.89% to 1.19%, indicating that the enriched search environment not only improved usability but also led to more successful purchase outcomes. In parallel, **search-driven revenue increased significantly**, from €8,680 to €11,550, highlighting the direct business impact of interface improvements.

Additionally, the **average order value (AOV) increased** from €310 to €330. This suggests that users were more likely to engage with higher-value products or complete more complex purchase journeys. This observation is consistent with previous studies emphasizing the role of visual cues and context-aware interfaces in promoting higher-value transactions.

Overall, the results provide strong empirical support for the hypothesis that enhancing search results pages with structured navigation, visual elements, and advanced filtering improves both user experience and business performance.

H₁ is therefore supported, as the enrichment of search results pages led to measurable improvements in behavioral and economic performance indicators.

Finally, the findings highlight that search effectiveness is not determined solely by algorithmic relevance. Even highly optimized ranking systems require well-designed user interfaces to guide user interaction effectively. The integration of filtering tools, visual elements, and structured information architecture plays a critical role in reducing search effort and improving overall system performance. These results reinforce the view that user experience in e-commerce search is inherently multidimensional, combining algorithmic, visual, and interaction design components.

4.2 Experiment 2: The Impact of Visual Banners on User Engagement and Search Performance

This experiment investigates the effect of visual content—specifically promotional banners—on user behavior and search performance. The analysis focuses on the high-frequency query “iPhone”, which represents one of the most frequently searched terms on the platform.

Μήνας	Queries (iPhone)	Banners' Clicks	CTR	CTR (%)	AOV (€)	Total Engagement (%)
Απρίλιος	10.264	0	3.489	33.6	365	33.6%
Μάιος	9.965	2.491	2.985 (+2.491)	29.9	387	44%

Two consecutive periods were compared. In the first period (April 2025), the search results page did not include any visual banners, and users were presented only with organic product listings. During this period, 10,264 queries generated 3,489 product clicks, corresponding to a product click-through rate (CTR) of 33.6%. The conversion rate reached 2.3%, while the average session duration on the search results page was 1 minute and 42 seconds.

In the second period (May 2025), a visually prominent promotional banner was introduced, while all other conditions remained constant. Although total queries slightly decreased to 9,965, user interaction increased significantly. Product clicks decreased to 2,985 (CTR: 29.9%), but the banner generated an additional 1,491 clicks. As a result, total engagement increased from 3,489 to 4,475 interactions, representing an increase of approximately 28%, with overall engagement reaching 44%.

Furthermore, session duration increased to 2 minutes and 27 seconds, indicating deeper user exploration. The conversion rate also improved, rising from 2.3% to 2.7%, while the average order value (AOV) increased from €365 to €387. These findings suggest that the presence of visual elements not only enhances engagement but may also guide users toward higher-value products or bundled offerings.

These results are consistent with prior research. Visual content has been shown to act as a stimulus that captures user attention and enhances purchase intention (Hu & Liu, 2021). Similarly, early-stage visual engagement plays a critical role in retaining users within the search environment (Ziakos & Vlachopoulou, 2023). The high interaction rate observed for the banner supports the argument that visual elements function as complementary navigation tools rather than substitutes for organic product listings.

Additionally, studies suggest that context-aware banners can improve perceived trust and guide user decision-making more effectively (Singh et al., 2023). The results of this experiment indicate that banners contribute to more structured and targeted navigation, potentially increasing both engagement and transaction value.

Overall, the findings support the hypothesis that visual enhancements improve both behavioral and business performance metrics.

H₂ is therefore supported, as the introduction of visual banners led to increased engagement, longer session duration, higher conversion rates, and improved revenue-related indicators.

4.3 Experiment 3: Preference for Generic Queries and the Broad-to-Narrow Search Behavior

This experiment investigates user preference for generic search queries and the role of information architecture in supporting broad-to-narrow navigation strategies.

The analysis is based on search data collected during the period from September 2, 2024, to May 31, 2025. Results indicate that approximately **76% of all queries were single-word or generic category-based terms** (e.g., “laptop”, “iPhone”, “refrigerator”, “Samsung”). This finding suggests that users do not initially seek highly specific products, but rather use the search function as a starting point for exploration.

This behavior aligns with the well-documented “broad-to-narrow” search strategy, where users begin with general queries and progressively refine results through filters, categories, or other navigational

tools. In this context, the search results page (SRP) plays a critical role—not only as a retrieval interface, but as a structured navigation environment.

The analysis further reveals that the effectiveness of this strategy is highly dependent on the design of the search interface. When the SRP provides rich refinement tools—such as filters, category tabs, and visual elements—users exhibit:

- lower refinement rates,
- higher conversion rates,
- and longer session durations.

Conversely, when such features are limited or absent, users tend to rely more heavily on repeated query reformulation, resulting in increased friction and reduced performance outcomes.

These findings are consistent with the “search-as-navigation” paradigm proposed by Teevan et al. (2005), which emphasizes the role of search systems as interactive navigation tools rather than purely algorithmic retrieval mechanisms.

Importantly, the results highlight that improvements in ranking algorithms alone (e.g., Learning to Rank or AI-based reranking) are insufficient to optimize search performance. Instead, the usability and structure of the search interface play an equally critical role in guiding users toward relevant results.

Overall, the findings support the hypothesis that users strongly prefer generic queries and rely on platform-provided navigation mechanisms to refine their search.

H₄ is therefore supported, as users demonstrate a consistent preference for general search terms combined with iterative refinement behavior.

4.4 Experiment 4: Emergence of Long-Tail Natural Language Queries and Implications for E-Commerce Search Systems

This experiment examines the emerging trend of long-tail queries formulated in natural language and its implications for the evolution of e-commerce search systems.

The analysis covers the period from September 2024 to May 2025 and reveals a gradual increase in the presence of natural language queries, such as “laptop for my son”, “TV for the bedroom”, or “laptop with good camera under €500”. Although the absolute volume of such queries remains limited—rising from 7 unique queries in September to 11 in May—the trend indicates a qualitative shift in how users express their search intent.

This shift appears to be influenced by the widespread adoption of AI-driven conversational systems, which have familiarized users with more natural and descriptive forms of interaction. As a result, users increasingly expect search systems to interpret intent rather than rely solely on keyword matching.

The findings are consistent with prior research suggesting that traditional keyword-based retrieval models are insufficient for handling natural language queries, as they fail to capture semantic meaning and user intent (Zhou et al., 2022). This limitation highlights the growing need for semantically aware search systems that incorporate technologies such as vector embeddings, natural language understanding (NLU), and context-aware matching.

Moreover, recent studies propose the integration of conversational refinement mechanisms and dynamic facet generation, enabling systems to decompose complex queries into structured attributes and guide users through an interactive search process (Ren & Li, 2025). For example, a query such as “gaming laptop with good cooling and long battery life” should not result in zero-results output, but rather be interpreted as a combination of constraints that can be translated into structured filters.

In the context of this study, the analysis focuses on the qualitative emergence of such queries rather than their quantitative performance (e.g., CTR or conversion rate). Nevertheless, their presence—even at a limited scale—signals a shift in user expectations that current keyword-based systems are not fully equipped to handle.

The examined search system, similar to many e-commerce platforms, remains primarily keyword-based and therefore demonstrates limited capability in processing natural language queries. This gap may negatively affect user experience, particularly for digitally native users who are accustomed to conversational AI interfaces.

Overall, the findings highlight an emerging challenge for e-commerce platforms: the need to transition from keyword-based search to more intelligent, intent-driven systems.

H₃ is partially supported, as the data indicate an emerging—but not yet dominant—shift toward natural language queries, with clear implications for future system design.

4.5 Interpretation of Results and Theoretical Implications

The findings of this study provide strong empirical evidence regarding user interaction with on-site search engines in e-commerce environments, while reinforcing key theoretical perspectives from the existing literature. Overall, the results highlight the importance of integrating algorithmic performance with user interface design, visual guidance, and emerging AI-driven capabilities.

First, the results confirm the critical role of search results page (SRP) design in shaping user behavior and system performance. The improvements observed in Experiment 1—particularly the significant reduction in refinement rate and the increase in conversion and revenue—support prior research suggesting that effective search systems must go beyond algorithmic ranking and incorporate navigational support mechanisms such as filters, structured categories, and guided interfaces (Shaban et al., 2021; Singh et al., 2023). These findings reinforce the concept that search functions as both a retrieval and navigation system.

Second, the results demonstrate the strong influence of visual content on user engagement and decision-making. As shown in Experiment 2, the introduction of visual banners significantly increased overall interaction levels, session duration, and conversion performance. This aligns with existing studies indicating that visual stimuli act as attention drivers, reduce cognitive load, and enhance user engagement (Hu & Liu, 2021; Ziakis & Vlachopoulou, 2023). The findings further suggest that visual elements operate as complementary navigation tools, guiding users through the decision-making process rather than competing with product listings.

Third, the dominance of generic queries and the prevalence of broad-to-narrow navigation patterns, as identified in Experiment 3, highlight the importance of information architecture and interface design in supporting exploratory search behavior. These findings are consistent with prior literature emphasizing that search systems should facilitate iterative refinement rather than assume precise initial queries (Lalitha et al., 2020; Zeebaree, 2023). This reinforces the “search-as-navigation” paradigm, where system usability plays a central role in search effectiveness.

Finally, the emerging presence of long-tail natural language queries, explored in Experiment 4, points to a gradual shift in user expectations driven by increased exposure to AI-based conversational systems. Although still limited in volume, these queries signal a transition toward more intent-driven search behavior. This finding supports recent research advocating for the integration of natural language understanding (NLU), semantic search, and context-aware systems in modern e-commerce platforms (Wen et al., 2023; Ren & Li, 2025).

Taken together, the results highlight the need for a holistic approach to search system design. Effective on-site search engines must combine:

- algorithmic relevance,
- user-centered interface design,
- visual and navigational support,
- and emerging AI-driven capabilities.

Focusing exclusively on ranking accuracy is insufficient to meet modern user expectations. Instead, search systems must evolve into integrated, interactive environments that support exploration, decision-making, and personalization.

5. Limitations

This study presents several limitations that should be acknowledged when interpreting the findings.

First, the analysis is based on data from a single e-commerce platform. Although the dataset originates from a large omnichannel retailer with high traffic volumes and extensive search activity, the results may still be influenced by platform-specific characteristics, market conditions, or brand-related factors. Consequently, the generalizability of the findings to other e-commerce contexts may be limited.

Second, the study primarily relies on quantitative data analysis. While large-scale behavioral data provide robust insights into user interaction patterns and system performance, they do not capture qualitative dimensions of user experience, such as perceived usability, trust, or emotional responses. The inclusion of qualitative methods—such as user interviews, usability testing, or heatmap analysis—could provide deeper insights into user motivations and decision-making processes.

Third, the analysis focuses exclusively on on-site search behavior and does not account for external factors that may influence user activity. Elements such as digital advertising, email marketing, push notifications, or offline interactions may affect both search queries and purchasing behavior, but are not incorporated into the current analysis.

Finally, although the dataset spans a nine-month period and includes a large volume of queries and sessions, the study captures behavior within a specific temporal window. Changes in market conditions, technological adoption, or user expectations over time may influence the observed patterns.

6. Conclusions

This study investigated the role and impact of on-site search engines within a large-scale e-commerce environment, focusing on how design, content, artificial intelligence, and user behavior influence search performance and user experience.

Drawing on real-world search data, experimental interventions, and theoretical grounding, the findings provide strong evidence that on-site search functions not merely as a technical feature, but as a strategic component of the digital customer journey. The results confirm that search performance is shaped not only by algorithmic relevance, but also by interface design, visual elements, and the availability of navigational support mechanisms.

Specifically, the study demonstrates that enhancing search results pages with structured elements—such as filters, tabs, and visual components—significantly improves both behavioral and business performance metrics, including refinement rate, conversion rate, and search-driven revenue. The introduction of visual content further amplifies user engagement and supports decision-making, highlighting the importance of integrating visual guidance into search environments.

In addition, the findings confirm that users predominantly rely on generic queries and adopt a broad-to-narrow search strategy, emphasizing the importance of information architecture and interactive navigation tools. At the same time, the gradual emergence of long-tail natural language queries suggests a shift toward more intent-driven search behavior, influenced by increasing familiarity with AI-based conversational systems.

Overall, the study highlights the need for a holistic approach to search system design. Effective e-commerce search engines must integrate:

- algorithmic accuracy,

- user-centered interface design,
- visual and navigational support,
- and AI-driven capabilities for intent understanding.

From a managerial perspective, the results suggest that investing in on-site search optimization can yield measurable improvements in both user experience and commercial performance. Organizations that prioritize search as a strategic asset—rather than a purely technical function—are better positioned to enhance customer engagement and achieve competitive advantage.

For future research, the integration of multichannel data and qualitative methods could provide deeper insights into user behavior. Furthermore, the adoption of natural language understanding (NLU) and conversational search systems represents a promising direction for addressing evolving user expectations.

In conclusion, this study demonstrates that optimizing the internal search experience can produce significant and measurable business impact. As user expectations continue to evolve, the role of on-site search will become increasingly central in shaping effective, personalized, and high-performing e-commerce environments.

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